

54.3 - Homework

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January 2026

Homework 3.3

7. Apply Cramers algorithm, and determine the values of the parameter s for which the system has a unique solution, and describe the solution.

$$6sx_1 + 4x_2 = 5, 9x_1 + 2sx_2 = -2$$

- Find $\det(A)$. Then find the values for s where the det will be 0

$$\det(A) = 12s^2 - 36$$

$$12(s^2 - 3) = 0$$

$$(s^2 - 3) = 0$$

$$s^2 = 3 \implies s = \pm\sqrt{3}$$

- Set up $A_1(b) \wedge A_2(b)$

$$A_1(b) = \begin{bmatrix} 5 & 4 \\ -2 & 2s \end{bmatrix} A_2(b) = \begin{bmatrix} 6s & 5 \\ 9 & -2 \end{bmatrix}$$

- Write x_1, x_2 . Note the textbook simplifies the answers.

$$x_1 = \frac{10s + 8}{12(s^2 - 3)}$$

$$x_2 = \frac{-12s - 45}{12(s^2 - 3)}$$

13. Compute the ad-jugate and use Theorem 8 to give the inverse

$$\begin{bmatrix} 3 & 5 & 4 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$$

- The adjugate matrix is the transpose of matrix cofactors
- Learned that you must apply the signs to your computed $\det(C_{ij})$. These signs follow a pattern of $+, -, +, -, +, -$

- Once you compute each $\det(C_{ij})$, the solution transpose is the form for A^{-1} . I.e., for this problem $i_2 = C_{21} = -1, C_{22} = -5, C_{23} = 7 \implies$

$$j_2 = \begin{bmatrix} -1 \\ -5 \\ 7 \end{bmatrix}$$

- Solution is:

$$A^{-1} = \frac{1}{6} \begin{bmatrix} -1 & -1 & 1 \\ 1 & -5 & 1 \\ 1 & 7 & -5 \end{bmatrix}$$

Where you scale the $\text{adj}(A)$ by 6 s.t. $(\forall a_{ij} \in A) \frac{a_{ij}}{6}$

21. Find the area of the parallelogram for said vertices: $(-2, 0), (0, 3), (1, 3), (-1, 0)$.

- Apply the **definition** that $\det(A) = \text{Area}$
- You can subtract points to make a vertex at the origin, in this case that would mean applying $+2\forall x_1 \in V$
- Definition of parallelogram, is you have 4 vertices that make up the area s.t. $\text{Area} = AB = DC \wedge BC = AD \wedge D = A + (C - B)$
- If you find $A, B, C \rightarrow D$ is forced
- Apply $\vec{u} = B - A \wedge \vec{v} = C - A \equiv$ two sides of a parallelogram.
- Thus, Area of a parallelogram is $\text{Area} = \det|(B - A)(C - A)|$
- No side effects proof:

$$A' = A + t, B' = B + t, C' = C + t \rightarrow$$

$$B' - A' = (B + t) - (A + t) = B - A \implies$$

Translation cancels and doesn't change det

- The solution in this case is

$$V = \begin{bmatrix} A + B & A + C \end{bmatrix} \rightarrow \det(V) = |-3|$$

Where $A + B = e_1$ and so forth.

29. Find formula for Area of a triangle with vertices $0, v_1, v_2 \in \mathbb{R}^2$

- Use geometry rules to apply $\frac{1}{2}[v_1 v_2]$ for $\frac{1}{2}b \cdot h$
- The above solution applies $\forall$, my solution only works for when x, y match up. I.e. $\frac{1}{2}[(x_1, x_2), (y_1, y_2)]$

35. Two parallelograms with same base and height have same area

- Vacuously True. This is analogous to $1 \cdot 1 = 1$

36. Applying a linear transformation to a region does not change its area

- False, because of the formula of a transformed area is $Area(T(S)) = |det(A)| \times Area(S)$. Meaning we must scale by a determinant.
- Counterexample is when it is a shearing transformation; because it has a $det(A) = 1$ and doesn't change the area.

37. If A is an invertible $n \times n$ matrix, then $A^{-1} = adj A$

- False, correct formula is:

$$A^{-1} = \frac{1}{det A} adj A$$

39. Test the inverse formula of Theorem 8 for a random 4 by 4 matrix A . Use your matrix program to compute the cofactors of 3 by 3 submatrices, construct the adjugate, and set $B = (adj A) / Det(A)$. Then compute $B \cdot inv(A)$, where $inv(A)$ is the inverse of A as computed by the matrix program. Use floating point arithmetic with the maximum possible number of decimal places

- The above question is essentially teasing that computer arithmetic creates floating integers that are really close to 0.

HW 4.1

1. Let V be the first quadrant in the xy -plane s.t.

$$V = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : x \geq 0, y \geq 0 \right\}$$

- If $\mathbf{u}$ and $\mathbf{v}$ are in V , is $\mathbf{u} + \mathbf{v}$ in V ?
- Find specific vector in $\mathbf{u}$ in V and a specific scalar c s.t. $c\mathbf{u}$ is NOT in V . It is enough to show that V is not in the vector space.

By definition, we know that addition is closed in V . Let $u = (x_1, y_1) \wedge v = (x_2, y_2) \rightarrow u + v = x_1 + x_2, y_1 + y_2 \in V \iff a_{ij} \in u, v \geq -1, \forall x, y$

A scalar that makes it not in a vector space is a **negative** c . Because we are set within the first quadrant and thus this will produce us outside of it.

Determine if given set is in the subspace of $\mathbb{P}_n$ for an appropriate value of n .

5. All polynomials of the form $\mathbf{p}(t) = at^2$ where $a \in \mathbb{R}$

Every coefficient in front of each basis vector is completely free. If even one coefficient is forced to be a specific number, **it is not a span**. ($a \in \mathbb{R})(S := at^2)[a \cdot (t^2) \implies \text{Span}(t^2)]$ Any span is automatically a subspace. To explicitly check we do:

- Let $p_1 = at^2 \wedge p_2 = bt^2 \rightarrow$

Proving closed under addition $p_1 + p_2 =$

$$at^2 + bt^2 = t^2(a + b) \implies ct^2 \in \text{Span}$$

Proving closed under multiplication $p \cdot at^2 =$

$$cp = c(at^2) = (ca)t^2 \implies dt^2 \in \text{Span}$$

Since a span is a subspace, you can just conclude there as well, $\text{Span}\{t^2\}$

9. Let H be the set of all vectors of the form $\begin{bmatrix} s \\ 3s \\ 2s \end{bmatrix}$. Find $\mathbf{v} \in \mathbb{R}^3$ s.t.

$H = \text{Span}(\mathbf{v})$. Why does this show H is a subspace of $\mathbb{R}^3$

- Logic behind writing things in parametric form to show subspaces
- $w = a_1v_1 + \dots + a_pv_p$ shows every vector in the span is just a linear combination

We will show that it can be written as column vectors to show linear combinations and thus it is a $\subseteq V$. Applied with Theorem 1,

$$\begin{bmatrix} s \\ 3s \\ 2s \end{bmatrix} \rightarrow s \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$$

13. Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$

- Is $\mathbf{w} \in \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$? how many vectors are in $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$?
- How many vectors are in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$?
- Is $\mathbf{w}$ in the subspace spanned by $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$? Why?

The vector $\mathbf{w}$ belongs to the $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} \iff \exists x_1, x_2, x_3, s.t.$

$$x_1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} + x_3 \begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

The vector equation is equivalent to a system of three linear equations in three unknowns. Set up the augmented matrix s.t. we have $A = [\mathbf{v}_1 \mathbf{v}_2 \mathbf{v}_3 | \mathbf{w}]$

When you solve, you will find that $\mathbf{w}$ is in the subspace spanned by the vectors $\iff$ the last row in $\mathbf{w}$ is all 0, because by echelon form we can see a free row and pivot.

Thus We have a total of 3 vectors, and $\{\mathbf{w} \in \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$

- (a) There are only three vectors in $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ and $\mathbf{w}$ is not of them
- (b) There are infinitely many vectors in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$
- (c) $\mathbf{w}$ is in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$
- (d) **NOTE:** $\mathbf{w} \in \text{Span} \iff Ax = w$ is consistent. In your reasoning, you said last row needs a free variable, that is not the case. Rather, it is more appropriate to say that row 3 needs to be consistent, i.e. $\mathbf{w}_3$ needs to be 0

25. Complete the following proof that the zero vector is unique. Suppose that $\mathbf{w} \in V$ has the property that $\mathbf{u} + \mathbf{w} = \mathbf{w} + \mathbf{u} = \mathbf{u} \quad \forall \mathbf{u} \in V$.

In particular, $0 + \mathbf{w} = 0$ BUT $0 + \mathbf{w} = \mathbf{w}$, by Axiom ??? Hence $\mathbf{w} = 0 + \mathbf{w} = 0$

By Axiom 4, that states $\exists \vec{0} \in V s.t. \mathbf{u} + \vec{0} = \mathbf{u}$

27. Fill in the missing axiom numbers in the following proof that $0\mathbf{u} = 0 \quad \forall \mathbf{u} \in V$

$$0\mathbf{u} = (0 + 0)\mathbf{u} = 0\mathbf{u} + 0\mathbf{u} \quad (a)$$

Add negative of $0\mathbf{u}$ to both sides:

$$0\mathbf{u} + (-0\mathbf{u}) = [0\mathbf{u} + 0\mathbf{u}] + (-0\mathbf{u})$$

$$0\mathbf{u} + (-0\mathbf{u}) = 0\mathbf{u} + [0\mathbf{u} + (-0\mathbf{u})] \quad (b)$$

$$\mathbf{0} = 0\mathbf{u} + 0 \quad (c)$$

$$\mathbf{0} = 0\mathbf{u} \quad (d)$$

My solution:

- a. Axiom 7
- b. Axiom 3
- c. Axiom 5
- d. Axiom 6

Textbook solution:

- a. Axiom 8
- b. Axiom 3
- c. Axiom 5
- d. Axiom 4

33. Given subspaces H and K of a vector space V , the sum of H and K is the set of all vectors $\in V$ that can be written as the sum of two vectors, one in H and the other in K ; that is,

$$H + K = (\mathbf{u}, \mathbf{v} \in H, K) \{w : w = \mathbf{u} + \mathbf{v}\}$$

- a. Show $H + K$ is a subspace of V
- b. Show that H is a subspace of $H + K$ and K is a subspace of $H + K$

To show subspaces, you must use the definition of what they are. I.e., the zero vector is within the subspaces, and operations are closed in addition and multiplication

By definition, we know that vectors $H \wedge K$ have to have their own $\vec{0}$. So we assume this to work with the following questions.

- a. Let $x_1 = (\mathbf{u}_1 + \mathbf{v}_1) \wedge x_2 = (\mathbf{u}_2 + \mathbf{v}_2)$

$$\mathbf{u} + \mathbf{v} = x_1 + x_2$$

Rearrange by like terms

$$(\mathbf{u}_1 + \mathbf{v}_1) + (\mathbf{u}_2 + \mathbf{v}_2) = (\mathbf{u}_1 + \mathbf{u}_2) + (\mathbf{v}_1 + \mathbf{v}_2)$$

Thus we have shown $\mathbf{u} \in H \wedge \mathbf{v} \in K$

Let $d = (\mathbf{u} + \mathbf{v})$. Let c be any scalar $\in \mathbb{R}$.

$$cd = c(\mathbf{u} + \mathbf{v})$$

$$c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$$

Thus we have shown $c\mathbf{u} \in H \wedge c\mathbf{v} \in K$ b. We can conclude that the vectors H, K have their respective zero vectors, are closed under scalar addition $\wedge$ multiplication $\implies H + K \subseteq V$. The same is assumed for $H \subseteq H + K \wedge K \subseteq H + K$.

4.2 HW

1. Determine if $\mathbf{w} = \begin{bmatrix} 1 \\ 3 \\ -4 \end{bmatrix} \in N(A)$, $A = \begin{bmatrix} 3 & -5 & -3 \\ 6 & -2 & 0 \\ -8 & 4 & 1 \end{bmatrix}$
- This problem is asking you to simply set up the equation as $A\mathbf{w} = 0$.
 - If its not equal to 0, it is **NOT** in the null space
 - Algorithm is $A\mathbf{w} = \sum A_{ij}x_i$ where $i = i$ for both $A \wedge x$

$$\begin{bmatrix} 3 + -15 + 12 \\ 6 + -6 + 0 \\ -8 + 12 + -4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Thus $A\mathbf{w} = 0$, we can conclude that $\mathbf{w} \in N(A)$

5. Find explicit description of $\text{Nul}(A)$ by listing vectors that span the null space

$$\begin{bmatrix} 1 & -2 & 0 & 4 & 0 \\ 0 & 0 & 1 & -9 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

General Solution:

- Express the null space in terms of a linear combination of vectors (parametric)
- **NOTE:** RREF when you can

$$\begin{cases} x_1 = 2x_2 - 4x_4 \\ x_2 = \text{free} \\ x_3 = 9x_4 \\ x_4 = \text{free} \\ x_5 = 0 \end{cases}$$

$$\rightarrow \text{Parametric sol: } x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -4 \\ 0 \\ 9 \\ 1 \\ 0 \end{bmatrix}$$

15. Find A s.t. given set is $\text{Col } A$

$$\left\{ \begin{bmatrix} 2s + 3t \\ r + s - 2t \\ 4r + s \\ 3r - s - t \end{bmatrix} : r, s, t \in \mathbb{R} \right\}$$

Solution for all cases: Write it in linear combination form. I.e. prove that the matrix A can be written as the Span of its columns to represent $\text{Col } A$

Let $W = \text{Col } A$

Direct Proof $A \equiv W$:

$W = x(a_1) + y(a_2) : x, y \in \mathbb{R} = \text{Span}\{a_1, a_2\} \implies$

Let $A = [a_1 a_2]$

Solution: Let $W = \text{Col } A \rightarrow A = W$:

$$r \begin{bmatrix} 0 \\ 1 \\ 4 \\ 3 \end{bmatrix} + s \begin{bmatrix} 2 \\ 1 \\ 1 \\ -1 \end{bmatrix} + t \begin{bmatrix} 3 \\ -2 \\ 0 \\ -1 \end{bmatrix}$$

$$\rightarrow \text{Span} \left\{ \begin{bmatrix} 0 \\ 1 \\ 4 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \\ 0 \\ -1 \end{bmatrix} \right\}$$

$$\implies A = \begin{bmatrix} 0 & 2 & 3 \\ 1 & 1 & -2 \\ 4 & 1 & 0 \\ 3 & -1 & -1 \end{bmatrix}$$

17. Find k s.t. $\text{Nul } A$ is a subspace of $\mathbb{R}^k$ AND find k s.t. $\text{Col } A$ is a subspace of $\mathbb{R}^k$

$$A = \begin{bmatrix} 2 & -6 \\ -1 & 3 \\ -4 & 12 \\ 3 & -9 \end{bmatrix}$$

- This problem is testing your ability to grapple with definitions
- E.g.: Let A be a $m \times n = 4 \times 3$ matrix. Then $\text{Nul}(A) \subseteq \mathbb{R}^3 \wedge \text{Col}(A) \subseteq \mathbb{R}^4$
- **Reason:** $A : \mathbb{R}^n \rightarrow \mathbb{R}^m, x \mapsto Ax$
 - Null space is $\forall x$ with $Ax = 0 \rightarrow$ where inputs live $\rightarrow \mathbb{R}^n$
 - Column space is all outputs $Ax \rightarrow$ lives where outputs live $\rightarrow \mathbb{R}^m$

- $N(A) \subseteq \mathbb{R}^2$
- $\text{Col}(A) \subseteq \mathbb{R}^4$
- **More hand wavy:** $N(A)$ is defined to be less than or equal to dimensions of j .

- **More hand wavy:** $\text{Col}(A)$ is defined to be less than or equal to dimensions of j

27. It can be shown that a solution of the system below is $x_1 = 3$, $x_2 = 2$, and $x_3 = -1$. Use this fact and theory from this section to explain why another solution $x_1 = 30$, $x_2 = 20$, $x_3 = -10$. (Observe how the solutions are related, but make no other calculations).

$$\begin{aligned}x_1 - 3x_2 - 3x_3 &= 0 \\ -2x_1 + 4x_2 + 2x_3 &= 0 \\ -x_1 + 5x_2 + 7x_3 &= 0\end{aligned}$$

- My reasoning is that if we can express this system of equations into an augmented matrix, then we can express it as a $\text{Col } A$. Thus by writing it in parametric form, we can show that there are scalars $\in \mathbb{R}$ and by separating scalars from the vectors creating a linear combination, we prove that there is some constant that is effecting the base vector $\forall a_n \in A$
- The correct way to word the above is express the solutions as an coefficient matrix. Since we know $Ax = 0 \rightarrow \forall c \in \mathbb{R} \quad (Acx) = c(Ax) = c(0) = 0$ by linearity.
- We can write the system of equations as a coefficient matrix, then we can write it as a linear combination.
- Furthermore, $\text{Nul } A$ is a subspace of $\mathbb{R}^3$ and closed under multiplication AND addition.

31. $T : \mathbb{P}_2 \rightarrow \mathbb{R}^2$ by $T(p) = \begin{bmatrix} \mathbf{p}(0) \\ \mathbf{p}(1) \end{bmatrix}$. If $\mathbf{p}(t) = 3 + 5t + 7t^2 \rightarrow T(p) = \begin{bmatrix} 3 \\ 15 \end{bmatrix}$.

- Show T is a linear transformation using polynomials $p, q \in \mathbb{P}_2$ (i.e. show $T(\mathbf{p} + \mathbf{q}) = T(\mathbf{p}) + T(\mathbf{q})$)
- Find a polynomial $\mathbf{p} \in \mathbb{P}_2$ that spans the kernel of T , and describe the range of T

35. Let V, W be vector spaces. Let $T : V \rightarrow W$ be linear trans. Let $U \subseteq V$ and $T(U)$ be the set of all images of the form $T(x)$ where $x \in U$. Show that $T(U) \subseteq W$

(1) Since $U \subseteq V \rightarrow \mathbf{0}_v \in U$, by linearity $T(\mathbf{0}_v) = \mathbf{0}_w$. So $\mathbf{0}_v = \mathbf{0}_w \implies \mathbf{0}_w \in T(U)$.

(2) Let $(T(\mathbf{x}), T(\mathbf{y}) \in T(U)) \rightarrow [T(\mathbf{x}) + T(\mathbf{y}) \in T(U)]$ and by $\mathbf{1} \rightarrow \mathbf{x} + \mathbf{y} \in U \implies T(\mathbf{x}) + T(\mathbf{y}) \in T(U)$. In other words since T is linear, then we proved its closed under addition.

Let $(\forall c \in \mathbb{R}) \rightarrow \mathbf{x} \in U \wedge U \subseteq V \rightarrow (c\mathbf{x} \in U)$ and by **2** we can show $T(c\mathbf{x}) = cT(\mathbf{x}) \wedge \in T(U)$. In other words $T(U)$ is closed under scalar multiplication and $T(U)$ is a subspace of W

4.3

Determine the basis for $\mathbb{R}^3$. If not bases find independence and which ones span $\mathbb{R}^3$ for questions **1-7**

1.

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

There is independence $\implies$ invertibility, and by *IVT* they span $\mathbb{R}^3$. In other words it is a basis.

3.

$$\begin{bmatrix} 1 & 3 & -3 \\ 0 & 2 & -5 \\ -2 & -4 & 1 \end{bmatrix}$$

In echelon form, we see that it spans $\mathbb{R}^2, \neg \mathbb{R}^3$ because there is dependence in row 3. In other words there is independence in $\mathbf{v}_1, \mathbf{v}_2$

$$\begin{bmatrix} 1 & 3 & -3 \\ 0 & 2 & -5 \\ 0 & 0 & 0 \end{bmatrix}$$

It is more correct to identify the pivot positions and then say that since there are two pivot positions, then this matrix is a 2-dimensional subspace of $\mathbb{R}^3$

$$Col(A) = \{x_1\mathbf{v}_1 + x_2\mathbf{v}_2 : x_1, x_2 \in \mathbb{R}\} \subseteq \mathbb{R}^3$$

5.

$$\begin{bmatrix} 1 & -2 & 0 & 0 \\ -3 & 9 & 0 & -3 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

REF(A) shows us that $Col(A) = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} \subseteq \mathbb{R}^4$.

$$\begin{bmatrix} 1 & -2 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

More appropriate to say $REF(A)$ shows us that $Col(A) = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \mathbb{R}^3$ because there is a pivot in every row.

We state that since there is a zero vector (dependence) present, that it cannot be a basis for $\mathbb{R}^3$

In other words, weve concluded that it is not a basis for $\mathbb{R}^3$ but it does span

7.

$$\begin{bmatrix} -2 & 6 \\ 3 & -1 \\ 0 & 5 \end{bmatrix}$$

Pivot $\in$ every column $\not\in$ every row. I.e. Free variable in Row 3 $\implies Col(A) \subseteq \mathbb{R}^3$, where $dim = 2$. Thus it is not a standard basis of $\mathbb{R}^3$.

The textbook says since it is $\neg \exists$ a pivot $\in$ every row, that means its columns cannot span $\mathbb{R}^3$. Same conclusion but worded differently.

9. Find bases for the null Spaces of matrix. (Example 3 from 4.2)

$$\begin{bmatrix} 1 & 0 & -3 & 2 \\ 0 & 1 & -5 & 4 \\ 3 & -2 & 1 & -2 \end{bmatrix}$$

$ref(A)$ and write in $x = x_p + x_{homogenous}$ We find

$$\begin{bmatrix} 1 & 0 & -3 & 2 \\ 0 & 1 & -5 & 4 \\ 0 & 0 & -2 & 0 \end{bmatrix} \rightarrow \begin{cases} x_1 = 3x_3 - 2x_4 \\ x_2 = 5x_3 - 4x_4 \\ x_3 = 0 \\ x_4 = free \end{cases} \rightarrow$$

$$x = \begin{bmatrix} 3x_3 - 2x_4 \\ 5x_3 - 4x_4 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 3 \\ 5 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -2 \\ -4 \\ 0 \\ 1 \end{bmatrix} = x_3 \vec{u} + x_4 \vec{v}$$

We showed that every linear combination of $\vec{u}, \vec{v} \subseteq Null(A)$. Thus basis for $Nul(A) =$

$$\left\{ \begin{bmatrix} 3 \\ 5 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ -4 \\ 0 \\ 1 \end{bmatrix} \right\}$$

11. Find a basis for the set of vectors in $\mathbb{R}^3$ in the plane $x + 2y + z = 0$. Hint think of the equations as a system of homogeneous equations.

On my first attempt of this problem, I wrote it into the homogenous solution incorrectly, s.t. there is 3 rows and 1 column. The correct approach is to write it as a coefficient matrix $\rightarrow$ express it in parametric form.

Write in augmented matrix form to identify pivot and free variables. Recognize that this is not a basis for $\mathbb{R}^3$. It can be a basis for $Nul(A)$:

$$A = \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}$$

Solve $Ax = 0$ general solution in terms of free variables:

$$x = -2y - z$$

Write in parametric form:

$$x = y \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \rightarrow \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

By rank nullity theorem we showed

$$\dim(Nul(A)) = j - \text{pivot} = 2$$

and instead of saying we showed $Col(A) \subset \mathbb{R}^3$ we say $Nul(A) \subset \mathbb{R}^3$

15. Find a basis for the space spanned by the given vectors

$$\begin{bmatrix} 1 \\ 0 \\ -3 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} -3 \\ -4 \\ 1 \\ 6 \end{bmatrix}, \begin{bmatrix} 1 \\ -3 \\ -8 \\ 7 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -6 \\ 9 \end{bmatrix}$$

For some reason I kept arithmetic slipping on $R4$. You make it into $RREF(A)$. x_3, x_4 should be free and that is what you take the span of

23. Suppose $\mathbb{R}^4 = \text{Span}\{v_1, \dots, v_4\}$. Explain why $\{v_1, \dots, v_4\}$ is a basis for $\mathbb{R}^n$.

There are two cases. We will have a linearly independence within $\{v_1, \dots, v_4\}$ or we will have dependence. In both cases, they are a basis for $\mathbb{R}^n$. This is because we Span the basis of $\mathbb{R}^n$ completely if we have n pivots, and if not we can remove v_k that makes it dependent s.t. we are given a basis for $Col A$

If $\text{Span}\{v_1, \dots, v_4\} = \mathbb{R}^4$ then with exactly 4 vectors in a 4d space they must be linearly independent, hence a basis. The issue with my solution is I showed it for the case of $\mathbb{R}^n$. Which I feel is a stronger proof.

23. **T/F** If $H = \text{Span}\{b_1, \dots, b_p\} \rightarrow \{b_1, \dots, b_p\}$ is a basis for H

Correct, if fully independent then we have a standard basis (i.e. full rank)

25. **T/F** The columns of an invertible $n \times n$ matrix form a basis for $\mathbb{R}^n$

Vacuously true in my opinion, because if we have the condition present that A is invertible then columns are linearly independent and span $\mathbb{R}^n$

Discs prep for Quiz

Disc6.3 Let T, U be linear trans. from $\mathbb{R}^n \rightarrow \mathbb{R}^n$ | $T(U(x)) = x \forall x \in \mathbb{R}^n$. Is it true that $U(T(x)) = x \forall x \in \mathbb{R}^n$? Why or why not.

We know that the domain is the same, I argued that since we are dealing with the same vector space $\forall x$ that this means $U(T(x)) = x \equiv T(U(x)) = x$

Let $A = T, B = U \rightarrow T(U(x)) = x \equiv ABx = x \implies AB = I$

WHY? Since AB is $n \times n \rightarrow$ if $\det(AB) \neq 0 \rightarrow AB = I \wedge (BA)^{-1} = AB$

Now, we can proceed with the assumption $B = A(B = A^{-1})$. $A = B(A = B^{-1})$. Since $A = B^{-1} \wedge B = A^{-1} \implies BA = I$ ■

WHY? For linear trans. a left inverse $TU = I$ is automatically a right inverse $UT = I$

Solution Guides to disc

- Disc8.1
- Disc6.3

Quiz 4

GSI stated problem 3.c or d is about the symmetrical skew

HW 4.4

1. Find $\mathbf{x}$ by coordinate vector $[\mathbf{x}]_{\mathcal{B}}$ and the given basis $\mathcal{B}$

$$\mathcal{B} = \left\{ \begin{bmatrix} 3 \\ -5 \end{bmatrix}, \begin{bmatrix} -4 \\ 6 \end{bmatrix} \right\}, [\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$
$$5(3, -5) + 3(-4, 6) = \begin{bmatrix} 3 \\ -7 \end{bmatrix}$$

The same process works for $\mathbb{R}^3$ in question 3. Too lazy to write it out.

5. Find the coordinate vector $[\mathbf{x}]_{\mathcal{B}}$ of $\mathbf{x}$ relative to the given basis $\mathcal{B} = \text{Span}\{b_1, \dots, b_n\}$
 $\mathbf{b}_1 = (1, -3), \mathbf{b}_2 = (2, -5), \mathbf{x} = (-2, 1)$

Since you can find the above question by **RREF**($\mathbf{Ax}$) OR A^{-1} times the vector $\mathbf{x}$:

$$\begin{bmatrix} 1 & 2 & -2 \\ -3 & -5 & 1 \end{bmatrix} \sim$$

$$\text{if } ad-bc \neq 0 \rightarrow A^{-1} = \frac{1}{ad-bc} \text{adj}(A) \text{ where } \frac{1}{\det(A)} \cdot \text{adj}(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \rightarrow$$

$$\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \cdot \begin{bmatrix} c_1 \\ c_n \end{bmatrix}$$

Thus, we set up the equation:

$$\begin{bmatrix} -5 & -2 \\ 3 & 1 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 + -2 \\ -6 + 1 \end{bmatrix} = \begin{bmatrix} 8 \\ -5 \end{bmatrix}$$

Again, you can get the same result by $rref(A)$ and checking $[x]_{\mathcal{B}}$ (i.e. the RHS b)

7.

11.

23. Let S be a finite set in a vector space V with the property that $\forall x \in V$ has a unique representation as a linear combination of elements of S . Show that S is a basis of V

S spans V if $\forall x \in V$ can be expressed as a linear combination of elements $\in S$. By definition, this means that $V = \text{Span}\{S\}$.

Now, to show independence we suppose that $(\forall v_i \in V) (\forall x \in V) [x = a_1v_1 + \dots + a_nv_n]$. Now let $x = b_1v_1 + \dots + b_nv_n$. Because we have the condition that x is a unique linear combination, then it must be the case that $a_i = b_i \forall i \implies (a_1 - b_1)v_1 + \dots + (a_n - b_n)v_n = 0 \implies \forall a_i - b_i = 0v_i = 0$. Which is the definition of linear independence. Thus we conclude that S is independent and spans $V \implies$ it is a basis.

25. Let $\mathcal{B} = (1, -4), (-2, 9)$. Since the coordinate mapping is determined by $\mathcal{B}$, it is a linear transformation from $\mathbb{R}^2 \rightarrow \mathbb{R}^2$, this mapping must be implemented by some 2×2 matrix A . Find it.

This question wants you to reverse engineer and find the coordinate vector. $P_{\mathcal{B}}^{-1} = \mathcal{B}$.

Apply inverse matrix formula to find $[(9, 2), (4, 1)]$

27. Let V be a vector space, $\mathcal{B}$ be a basis from $\mathcal{B}_1, \dots, \mathcal{B}_n$, and the coordinate mapping $\mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$. Show that the coordinate mapping is one-to-one.

Suppose $\forall \mathbf{u}, \mathbf{w} \in V, [\mathbf{u}]_{\mathcal{B}} = [\mathbf{w}]_{\mathcal{B}} \implies \exists$ specific $(c_1, \dots, c_n)$ s.t. $\sum c_i b_i = \mathbf{u} \wedge \mathbf{w}$. By this condition we can say that

$$\mathbf{u} = \mathbf{w} = c_1 b_1 + \dots + c_n b_n$$

$\rightarrow$ Thus, the coordinate vector is one to one.

Discussion 9

1. In the vectorspace of all real functions, find a basis for the spaced spanned by $\{\sin(t), \sin(2t), \sin(t)\cos(t)\}$

The solution for the above is to simply create the $Col(A)$ that will be a basis. In this case its $\{sint, sintcost\}$ due to the double angle formula that creates dependence.

2. Suppose $T : V \rightarrow W$ is one to one, s.t. $T(v) = T(u) \implies v = u$. Show that if $\{v_1, \dots, v_n\}$ are independent, $\rightarrow$ so are the vectors of $\{T(v_1), \dots, T(v_n)\}$

Initial thoughts: - I think independence is held in the transformations. Not sure how to prove that with algebra.

Can argue with injectivity and surjectivity

Source: <https://math.stackexchange.com/questions/1198053/prove-that-tv-1-tv-n-is-a-basis-for-w>

You must say not all zero, when writing $c_1 \text{ to } c_n \text{ for } T(v_1) - \dots - T(v_n)$

3.

[Change of basis formulas](#) Relevant reading $T(e_i) = b_i \hat{v} = P_B[v]_B$

Change of basis matrix is the $n \times n$: $n \times n$ has to be $b_1 \text{ to } b_n [n \times n] [v_1 \text{ to } v_n]_B = [v_1 \text{ to } v_n]$

First dont assume $n \times n$ is $b_1 \text{ to } b_n$, rather choose $c_1 \text{ to } c_n$ to then prove?

$n \times n$ has to be invertible and contain properties of IMT to produce a unique solution

$$v = v_1 \text{ to } v_n e_n v_B v_1 b_1 \text{ to } v_n b_1$$

b_1 rewrite them in the e^1 basis

$$T[V]_B \text{ to } V \text{ is } e_1 \text{ to } b_i \text{ meaning } P_B = [b_1 | \dots | b_n]$$

Notes and Resources I feel are valuable

- [Linear Independence and Spanning](#)
 - [Discrete 10 Problem a.](#)
 - [Injective and Surjective](#)
 - [Inverse Math Aid kit](#)
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Discussion 10

Say $w = a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4$, then if $\exists$ dependence this implies $\exists c_{1-4} \in \mathbb{R}$ s.t. $c_1 v_1 + c_2 v_2 + c_3 v_3 + c_4 v_4 = \{\vec{0}\}$ with the condition that c_i are not all zero.

Assume dependence exists, then we must choose a $v_i \neq 0$ to be dependent, then it can be rewritten as another expression for w . Take the example for v_4 :

$$w = a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4$$

$$w = a_1 v_1 + \dots + a_4 (c_1 v_1 + c_2 v_2 + c_3 v_3)$$

$$w = (a_1 + a_4 c_1) v_1 + (a_2 + a_4 c_2) v_2 + (a_3 + a_4 c_3) v_3$$

if $\{p_1, \dots, p_n\} \wedge [q]_{\mathcal{B}} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$ given, then $a_1(p_1) + \dots + a_n(p_n)$

In other words, express as a linear combination and group like terms to find $q(t)$ when we have $\{p_1, \dots, p_n\} \wedge [q]_{\mathcal{B}}$

Quiz 10

3. You are given vectors $\{v_1, v_2, v_3\} \in \mathbb{R}^2$

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, v_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, v_3 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$\exists$ dependence within these vectors. Write two ways that $v = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ can be expressed.

$$\begin{bmatrix} 3 \\ 4 \end{bmatrix} = v_1 + v_3$$

4.6

1. Let $B = \{\mathbf{b}_1, \mathbf{b}_2\}, C = \{\mathbf{c}_1, \mathbf{c}_2\}$ be bases for vector space V , suppose $\mathbf{b}_1 = 6\mathbf{c}_1 - 2\mathbf{c}_2, \mathbf{b}_2 = 9\mathbf{c}_1 - 4\mathbf{c}_2$

1. Find change of coord matix from $B \rightarrow C$

2. Find $[x]_C, \mathbf{x} = -3\mathbf{b}_1 + 2\mathbf{b}_2$ with part a.

To find $P_B \leftarrow_C$:

$$\begin{bmatrix} 6 & 9 \\ -2 & -4 \end{bmatrix}$$

Then, use the change of basis to represent the vectors within $\mathbf{x}$ as $\mathbf{x} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$

$$\begin{bmatrix} 6 & 9 \\ -2 & -4 \end{bmatrix} \begin{bmatrix} -3 \\ 2 \end{bmatrix} = \begin{bmatrix} 18 - 18 \\ 6 - 8 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$$

3. Let $U = \{u_1, u_2\}, W = \{w_1, w_2\}$ be bases for V , and let $P = \{[u_1]_W, [u_2]_W\}$. What is satisfied by $P \forall x \in V$? [Solution](#)

$$[x]_U = P[x]_W \vee [x]_W = P[x]_U$$

Because $P_W \leftarrow_U$, then P changes the basis from $U \rightarrow V$ Meaning $[x]_W = P[x]_U$.

$[x]_U = P[x]_W \implies P$ changes basis from $W \rightarrow U$, which is not the case.